

Golam Ishtiaq

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Education

Nanyang Technological University (NTU), Singapore

Expected Graduation: 2028

Bachelor of Science in Physics, Second Major in Data Analytics

GPA: 4.88/5.00

Relevant Coursework: Linear Algebra, Statistical Mechanics, Quantum Mechanics 2, Quantum Information, Introductory General Relativity, Data Structure and Algorithms

Kushtia Government College, Bangladesh

Graduated: 2023

Higher Secondary Certificate (Bangladesh National Curriculum)

GPA: 5.00/5.00

Achievements

International Physics Olympiad: Awarded Bronze Medal in 2023 and 2022 (top 15% worldwide)

Asian Physics Olympiad: Awarded Honorable mention in 2022 (top 20% of participants worldwide)

Nanyang Scholarship Recipient: Received full scholarship awarded to top 1% of students at NTU.

UK Integration Bee Finalist (Singapore Round): Top 10 finalist in 2025 and 2026.

PLANCKS Competition (Singapore Round, 2026): Runner-up

Key Projects

Review of Gauge Field Theories

Odyssey Research Project, May 2026 – Present

- Undertaking a guided review of quantum mechanical formalism and relativistic matter fields (Klein-Gordon and Dirac).
- Studying the theoretical formulation of Abelian and non-Abelian gauge field theories.

Thermalisation in 1D Ultracold Gases

URECA Research, Aug 2025 – Present

- Analyzed 1D gas dynamics using the Lieb-Liniger model, applying third-order perturbation theory to solve for two-point phase-phase correlations and capture non-linear dynamics.
- Truncated the correlation hierarchy by applying Gaussian state approximation and obtained an approximate closed-form equation for two-point phase-phase correlations.
- Analyzing simulated thermalization data to track the time-evolution of momentum mode correlation decay, mutual information, and logarithmic negativity.
- Characterizing the transient emergence and subsequent decay of non-Gaussianity during the relaxation process between known Gaussian initial and final states.

Image Processing Algorithms (Python)

Academic Coursework Project

- Developed a command-line interface engine for real-time image matrix manipulation.
- Implemented core algorithms (Gaussian blur, contrast adjustment, grayscale conversion) from scratch to optimise processing efficiency, utilising NumPy for vectorized array operations.

Experience

Physics Team Lead (Olympiad Section) & Evaluator, Mercor Intelligence

Remote, Jul 2025 – Jan 2026

- Selected to lead a 15-member physics team within a broader group of 50+ subject-matter experts developing high-quality training data for a confidential top-3 global GenAI model.
- Designed adversarial, Olympiad-level physics problems to probe model reasoning failures and improve evaluation robustness.
- Reviewed and curated high-accuracy solutions and annotations for reinforcement learning and supervised training workflows.

Student Ambassador, NTU SPMS and Admissions

Feb 2026 – Present

- Assisted the Office of Admissions during engagement sessions for accepted Olympiad medalists, providing guidance through the university acceptance process.
- Offered 1-on-1 follow-up support to prospective students, answering individual enrollment inquiries and sharing personal experiences at NTU.
- Supported official SPMS scholarship interview sessions, accompanying candidates through the evaluation process, and discussing academic life.
- Served as a designated “Scholarship Buddy” for incoming SPMS scholars, providing personalized mentorship on coursework navigation, academic planning, and campus life.

Physics Olympiad Tutor

Freelance

- Mentored 15+ high school students for Olympiad competitions; several students successfully advanced to national rounds.
- Designed and delivered an advanced physics curriculum focusing on problem-solving strategies.

Academic Tutor & Instructor

Volunteer

- Prepared 50+ students for competitive entrance exams with personalised strategies in physics and mathematics.
- Served as a volunteer for the Bangladesh Physics Olympiad, setting competition problems and training students at national camps.
- Conducted science outreach sessions on YouTube, reaching over 1.5k views to promote STEM education.

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib), C, C++, R.

Tools: LaTeX, Mathematica.

Other: Mathematical Modeling, Algorithm Design, Technical Writing.

Languages

Fluent in English and Bengali.

Hobbies and Interests

Competitive programming, Active contributor on Physics Stack Exchange.